Zero Trust Network Access (ZTNA) with Zscaler: Securing User-to-Application Connectivity

GROUP DETAILS

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PROJECT DETAILS

Title:

Zero Trust Network Access (ZTNA) with Zscaler: Securing User-to-Application Connectivity

INTRODUCTION:

In today's digital world, cyber threats are increasing rapidly, making traditional security models less effective. Organizations that rely on perimeter-based security, such as firewalls and Virtual Private Networks (VPNs), are vulnerable to attacks like unauthorized access, data breaches, and lateral movement within networks. These traditional systems often grant broad access once a user is authenticated, which creates significant security risks. Therefore, there is a growing need for a more advanced and secure approach to protect sensitive data and applications.

This project focuses on identifying and analyzing vulnerabilities in web applications and implementing a modern security framework known as Zero Trust Network Access (ZTNA). In the first stage, common web application vulnerabilities are studied based on OWASP standards. In the second stage, Nessus, a powerful vulnerability scanning tool, is used to detect real-time security issues such as outdated protocols, missing security headers, and weak configurations. These vulnerabilities are analyzed in detail along with their impact and recommended solutions.

In the final stage, the project introduces ZTNA using Zscaler, which follows the principle of "never trust, always verify." Unlike traditional VPNs, ZTNA provides secure, identity-based access only to specific applications rather than the entire network. By continuously verifying user identity, device health, and access policies, this approach significantly reduces the attack surface and enhances overall security. This project demonstrates how combining vulnerability assessment with Zero Trust architecture can create a robust and scalable cybersecurity solution.

ABSTRACT:

The increasing complexity of cyber threats and the limitations of traditional network security models have made it essential to adopt more advanced and reliable security frameworks. Conventional approaches such as Virtual Private Networks (VPNs) provide broad network access once a user is authenticated, which increases the risk of unauthorized access and lateral movement within the network. This project aims to address these challenges by combining vulnerability assessment techniques with Zero Trust Network Access (ZTNA) to enhance overall system security.

The project is divided into three main stages. In Stage 1, web application vulnerabilities are identified and analyzed based on OWASP Top 10 standards, focusing on issues such as improper input validation, missing security headers, and outdated protocols. Stage 2 involves performing vulnerability scanning using the Nessus tool, which detects real-time security weaknesses and provides detailed reports including severity levels, risk impacts, and remediation steps. These findings help in understanding potential threats and improving system security.

In Stage 3, the project introduces Zero Trust Network Access using Zscaler, which follows the principle of “never trust, always verify.” This approach ensures that users are granted access only to specific applications after verifying their identity and device security posture. By eliminating implicit trust and enforcing strict access control policies, ZTNA significantly reduces the attack surface and prevents unauthorized access. The project demonstrates how integrating vulnerability assessment with Zero Trust architecture can create a scalable, efficient, and secure environment for modern network infrastructures.

USE CASE SCENARIOS

1. Secure Remote Access for Faculty to ERP System

A faculty member needs to access the university ERP system from home to upload student data. Using ZTNA with Zscaler, the user is authenticated through an identity provider with multi-factor authentication (MFA). The device is also checked for security compliance before access is granted.

Output:

User identity successfully verified

Device posture validated (antivirus active, updated OS)

Secure connection established only to ERP application

Internal network remains hidden from the user

2. External Researcher Accessing Database

An external researcher requires temporary access to a specific research database hosted on the institution's server. The administrator creates a time-limited access policy using Zscaler.

Output:

External user authenticated using temporary credentials

Access restricted to only the research database

No visibility of other internal resources

Access automatically revoked after expiration

3. Student Access to Online Examination Portal

Students need to access an internal online examination portal during exams. ZTNA ensures that only authenticated users with secure devices can access the portal.

Output:

Large number of users authenticated simultaneously

Access granted only to exam application

Unauthorized access to other systems blocked

System remains secure under high traffic

🔹 4. IT Administrator Performing Server Maintenance

An IT administrator needs to update an internal server. Instead of full network access, ZTNA provides restricted access only to the required server interface.

Output:

Admin identity verified using MFA

Access limited to specific server management interface

All actions logged for auditing

Unauthorized access attempts blocked

components.

🔹 5. Temporary Contractor Access

A contractor is hired for a short-term project and needs access to a file server. ZTNA allows controlled and time-bound access without adding the contractor to the main network.

Output:

Contractor identity verified through secure authentication

Access granted only to required file server

All activities monitored and logged

Access automatically revoked after project completion

Stage-1

Title:

Zero Trust Network Access (ZTNA) with Zscaler: Securing User-to-Application Connectivity

OVERVIEW

This stage of the project focuses on identifying and analyzing common web application vulnerabilities using standard security practices such as the OWASP Top 10. Web applications are often exposed to various threats due to improper coding, misconfigurations, and outdated components. These vulnerabilities can lead to serious security issues such as unauthorized access, data leakage, and system compromise. Therefore, understanding and detecting these vulnerabilities is an essential step in strengthening overall cybersecurity.

In this stage, different types of vulnerabilities such as improper input validation, missing security headers, weak encryption protocols, and denial-of-service conditions are studied. Each vulnerability is examined in detail with its corresponding CWE (Common Weakness Enumeration) number, description, business impact, and recommended solution. This helps in understanding how attackers exploit these weaknesses and how they can be mitigated effectively.

The main objective of this stage is to create awareness about web security risks and to provide a structured approach to identifying and resolving vulnerabilities. By analyzing these issues, organizations can improve their application security, reduce attack surfaces, and ensure better protection of sensitive data and systems.

Vulnerability Table

S. No	Vulnerability Name	CWE No
1	TLS (Multiple Issues)	326
2	SSL (Multiple Issues)	327
3	HTTP (Multiple Issues)	319
4	IETF MD5 (Multiple Issues)	327
5	Service Detection	200
6	OS Identification	200
7	Additional DNS Hostnames	200
8	Nessus SYN scanner	200

📄 Vulnerability Report

1. Vulnerability Name: TLS (Multiple Issues)

CWE: 326

Description:

The targetsystemsupports outdated and deprecated versionsof the Transport Layer Security (TLS) protocol, including TLS 1.0 and TLS 1.1. These older protocol versions have known cryptographic weaknesses and are no longer considered secure.

Modern security standards recommend using TLS 1.2 or higher, as earlier versions are vulnerable to various attacks such as downgrade attacks, man-in-the-middle (MITM) attacks, and exploitation of weak cipher suites. The presence of these deprecated protocols indicates improper security configuration in the server.

Business Impact:

Theuse of outdated TLS versions can expose sensitive data transmitted between clients and the server to attackers. This may lead to:

Data interception and leakage

Unauthorized access to confidential information

Increased risk of man-in-the-middle attacks

Non-compliance with security standards (PCI-DSS, GDPR, etc.)

Damagetoorganizationalreputationand trust

Vulnerability Path: https://localhost

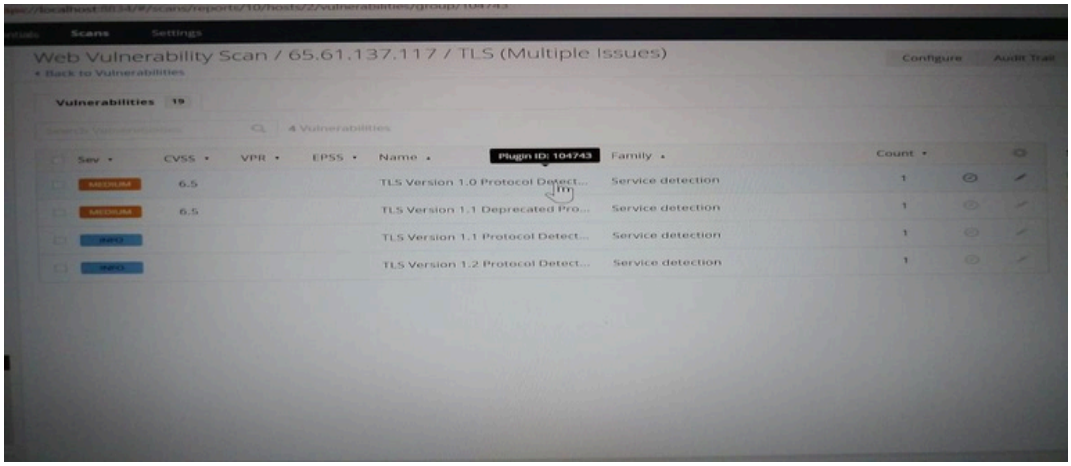
Steps to Reproduce:

➤ Perform a vulnerability scan using Nessus on the target system.

➤ Navigate to the “TLS (Multiple Issues)” section in the scan results.

➤ Observe that the server supports deprecated protocols such as TLS 1.0 and TLS 1.1.

➤ Verify that the server accepts connections using these insecure protocol versions.



Recommendation:

• DisablesupportforTLS 1.0 and TLS 1.1 on the server.

• Configure the system to use only secure protocols such as TLS 1.2 and TLS 1.3.

• Update server configurations and cryptographic libraries to the latest versions.

• Use strong cipher suites that provide forward secrecy.

• Regularly perform security scans to ensure compliance with modern security standards.

2. Vulnerability Name: SSL (Multiple Issues)

CWE: 326

Description:

The target system is configured with multiple SSL-related issues, including weak cipher suites, improper encryption configurations, and exposure of SSL certificate details. The scan results indicate that the server supports cipher block chaining (CBC) mode ciphers and potentially weak or outdated encryption algorithms.

Additionally, SSL certificate information is publicly accessible, which may reveal sensitive metadata about the server. Improper SSL configurations can weaken the overall security of encrypted communications and make the system vulnerable to cryptographic attacks.

Business Impact:

Improper SSL configuration can lead to several security risks, including:

Exposure of sensitive data through weak encryption

Increased risk of man-in-the-middle (MITM) attacks

Exploitation of cryptographic weaknesses (e.g., CBC attacks like BEAST)

Leakage of server or certificate information

Loss of user trust and compliance violations

Vulnerability Path: https://localhost

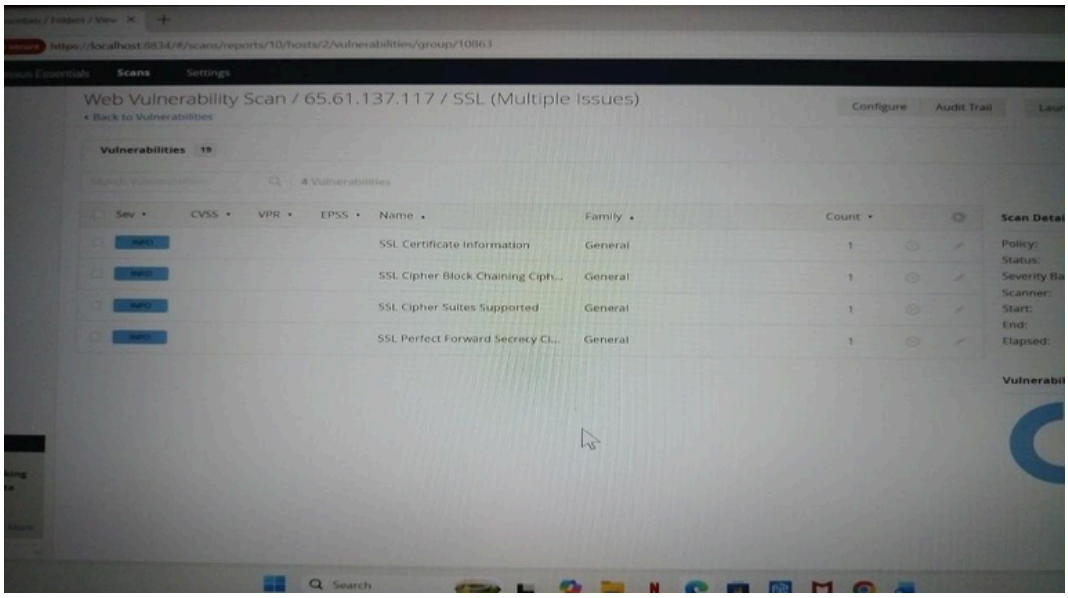
Steps to Reproduce:

➤ Perform a Nessus vulnerability scan on the target system.

➤ Navigate to the “SSL (Multiple Issues)” section in the scan report.

➤ Observe supported cipher suites, certificate details, and encryption configurations.

➤ Identify weak cipher suites (e.g., CBC mode) and exposed SSL certificate information.



Recommendation:

•Disable weak and outdated cipher suites (especially CBC-based ciphers).

•Configure the server to use strong encryption algorithms such as AES-GCM.

•Ensure proper SSL/TLS configuration following industry best practices.

•Restrict unnecessary exposure of SSL certificate information.

•Regularly update SSL/TLS libraries and perform security audits.

3. Vulnerability Name: HTTP (Multiple Issues)

CWE-200 (Information Exposure)

Description:

The target web server exposes multiple HTTP-related issues, including disclosure of server information, supported HTTP methods, and lack of secure communication enforcement.

The scan reveals that the server responds with detailed information such as HTTP version, server type (e.g., Apache), and supported methods (GET, POST, OPTIONS, etc.). Additionally, the server does not enforce secure communication using HTTPS and lacks HTTP Strict Transport Security (HSTS), which makes it vulnerable to downgrade attacks.

These misconfigurations allow attackers to gather information about the server and potentially exploit known vulnerabilities.

Business Impact:

The presence of HTTP-related issues can lead to:

Information leakage about server configuration

Increased attack surface for attackers

Vulnerability to man-in-the-middle (MITM) attacks

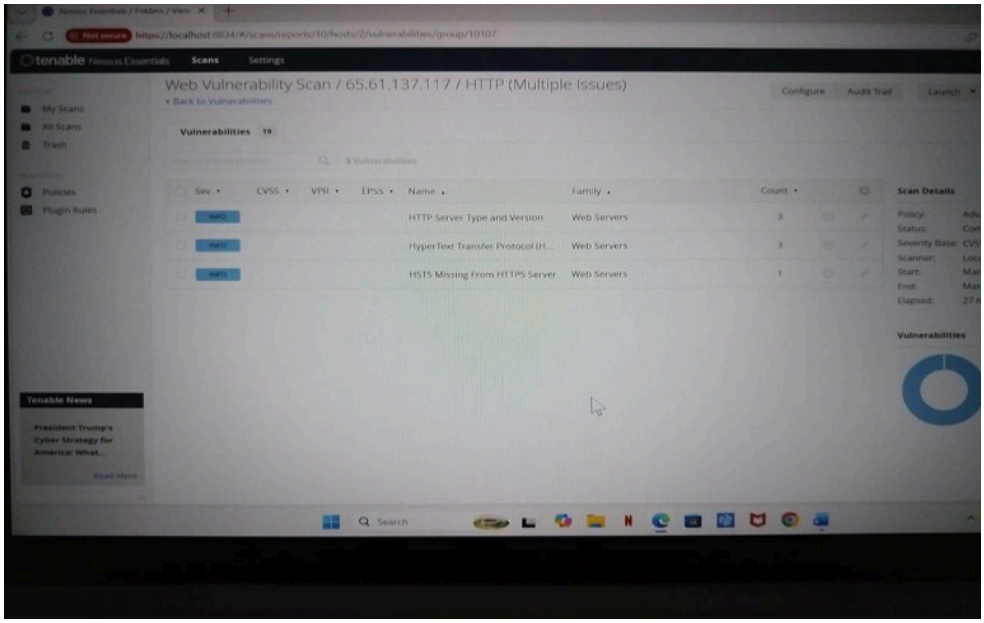
Exposure to downgrade attacks (HTTP instead of HTTPS)

Risk of unauthorized access and exploitation

Vulnerability Path: http://localhost (as identified in Nessus scan)

Steps to Reproduce:

- Perform a Nessus scan on the target system.
- Navigate to the “HTTP (Multiple Issues)” section.
- Observe server response headers showing server type and version.
- Identify supported HTTP methods and protocol details.
- Verify absence of HSTS header and lack of HTTPS enforcement.



Recommendation:

•Disableunnecessary HTTPmethods (e.g., OPTIONS, TRACE if enabled).

•Hide server version and sensitive information from HTTP headers.

•Enforce HTTPS across the application.

•Implement HTTP Strict Transport Security (HSTS).

•Regularly update web server software to the latest version.

4. Vulnerability Name: IETF MD5 (Multiple Issues)

CWE-327 (Use of a Broken or Risky Cryptographic Algorithm)

Description:

The targetsystem uses SSL certificates signed with weak hashing algorithms such as MD5 or SHA-1. These hashing algorithms are considered cryptographically broken and vulnerable to collision attacks.

An attacker can exploit these weaknesses to create fraudulent certificates or impersonate a legitimate server. This compromises the trust model of SSL/TLS and weakens secure communications.

Modern security standards strongly discourage the use of MD5 and recommend stronger hashing algorithms such as SHA-256 or higher.

Business Impact:

The use of weak hashing algorithms can lead to:

Certificate forgery and impersonation attacks

Man-in-the-middle (MITM) attacks

Loss of data confidentiality and integrity

Compromise of secure communication channels

Non-compliance with security standards

Vulnerability Path:https://localhost (as identified in Nessus scan)

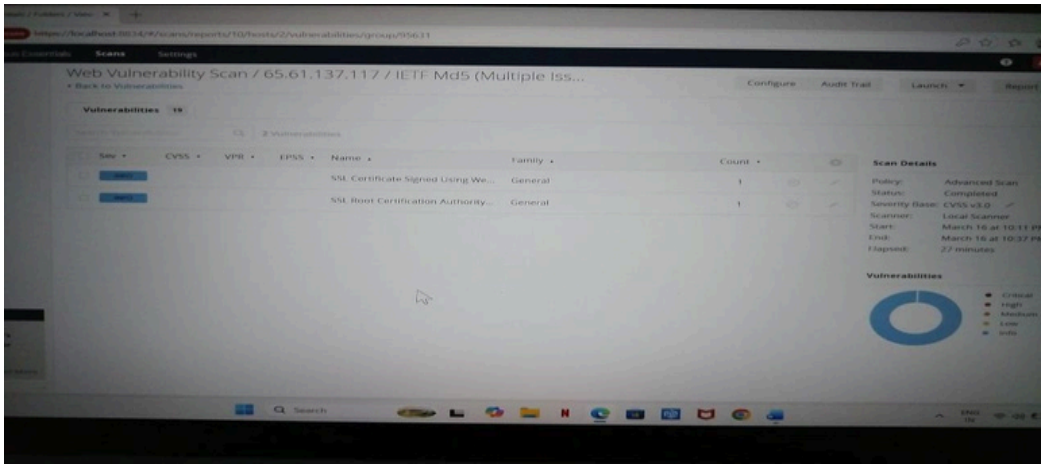
Steps to Reproduce:

➤ Perform a Nessus vulnerability scan on the target system.

➤ Navigate to the “IETF MD5 (Multiple Issues)” section.

➤ Observe that SSL certificates are signed using weak hashing algorithms (MD5/SHA-1).

➤ Verify certificate details showing insecure signature algorithms.



Recommendation:

•Replacecertificatessignedwith MD5 or SHA-1 with stronger algorithms like SHA-256 or SHA-3.

•Obtain certificates from trusted Certificate Authorities (CA).

•Regularly update SSL/TLS configurations and certificates.

•Disable support for weak hashing algorithms in the system.

5. Vulnerability Name: Service Detection

CWE: 200 (Information Exposure)

Description:

The vulnerabilityscanner was able to identify running services on the target system by analyzing service banners or responses. This means the server reveals information about the services running on specific ports, such as web servers or other network services.

Service detection itself is not a direct vulnerability, but it provides attackers with useful information about the system. This information can be used to identify

potential entrypoints and known vulnerabilities associated with detected services.

Business Impact:

Exposure ofserviceinformation can lead to:

Increased attack surface for attackers

Easier identification of exploitable services

Targeted attacks based on known vulnerabilities

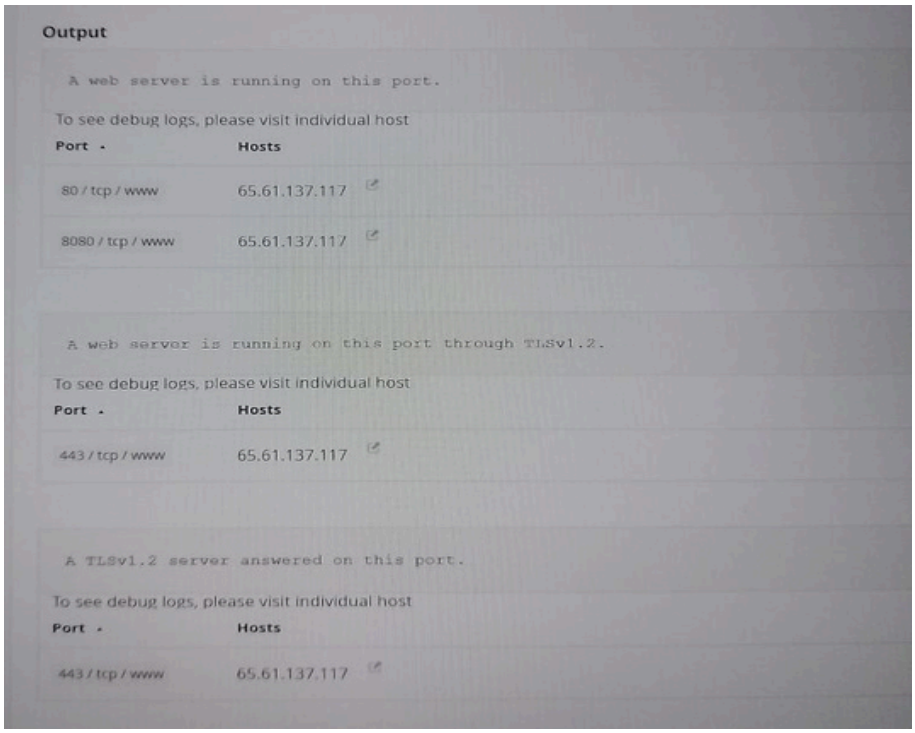
Information gathering for further exploitation

Vulnerability Path:https://localhost (as identified in Nessus scan)

Steps to Reproduce:

➤ Perform a Nessus scan on the target system.

- Navigate to the “Service Detection” section.
- Observe that services running on specific ports are identified.
- Verify that service banners or responses reveal system information.



Output

A web server is running on this port.

To see debug logs, please visit individual host

Port	Hosts
80 / tcp / www	65.61.137.117
8080 / tcp / www	65.61.137.117

A web server is running on this port through TLSv1.2.

To see debug logs, please visit individual host

Port	Hosts
443 / tcp / www	65.61.137.117

A TLSv1.2 server answered on this port.

To see debug logs, please visit individual host

Port	Hosts
443 / tcp / www	65.61.137.117

Recommendation:

•Disable unnecessary services running on open ports.

•Restrict access to services using firewall rules.

•Configure services to hide orminimize banner information.

•Regularly monitor and auditexposed services.

6. Vulnerability Name: OS Identification

CWE: 200 (Information Exposure)

Description:

Thevulnerability scanner was able to identify the operating system running on the target machine by analyzing network responses such as TCP/IP stack behavior, HTTP headers, or other fingerprinting techniques.

This means the system reveals information about its underlying operating system, which can be used by attackers to tailor specific attacks based on known

vulnerabilities ofthat OS.

Business Impact:

Exposure ofoperating system information can lead to:

Targeted attacks based on OS-specific vulnerabilities

Increased risk of exploitation using known exploits

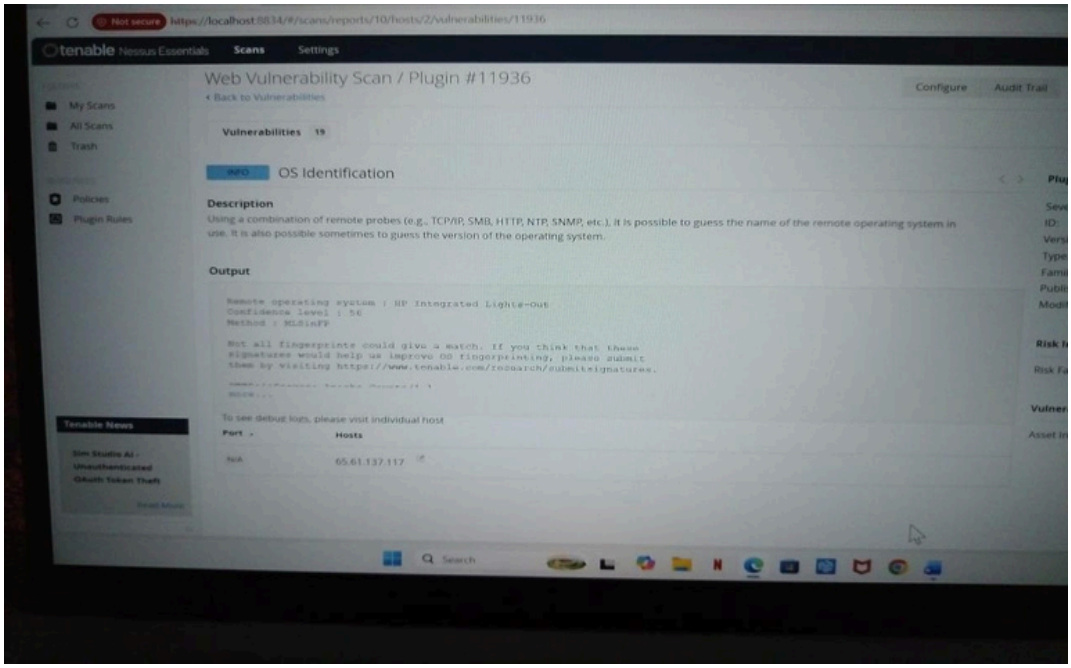
Easier reconnaissance for attackers

Reduced overall system security

VulnerabilityPath:https://localhost (as identified in Nessus scan)

Steps to Reproduce:

- Perform a Nessus scan on the target system.
- Navigate to the “OS Identification” section.
- Observe that the scanner identifies the operating system details.
- Verify that system responses reveal OS-related information.



Recommendation:

•Disableunnecessary service banners and OS fingerprinting exposure.

•Configure firewalls to limit external access to system information.

•Use intrusion detection/prevention systems (IDS/IPS).

•Regularly update and patch the operating system.

7. Vulnerability Name: Additional DNS Hostnames

CWE: 200 (Information Exposure)

Description

The vulnerability scan identified additional DNS hostnames associated with the target system. These hostnames are different from the primary hostname but resolve to the same server.

This occurs when multiple domains or subdomains are configured on the same web server (virtual hosting). Attackers can use this information to discover hidden or less secure services that are not intended for public access.

This exposure aids attackers in the reconnaissance phase by providing more entry points to analyze and exploit.

Business Impact:

The presence of additional DNS hostnames can lead to:

Discovery of hidden or internal services

Increased attack surface

Exposure of unpatched or insecure subdomains

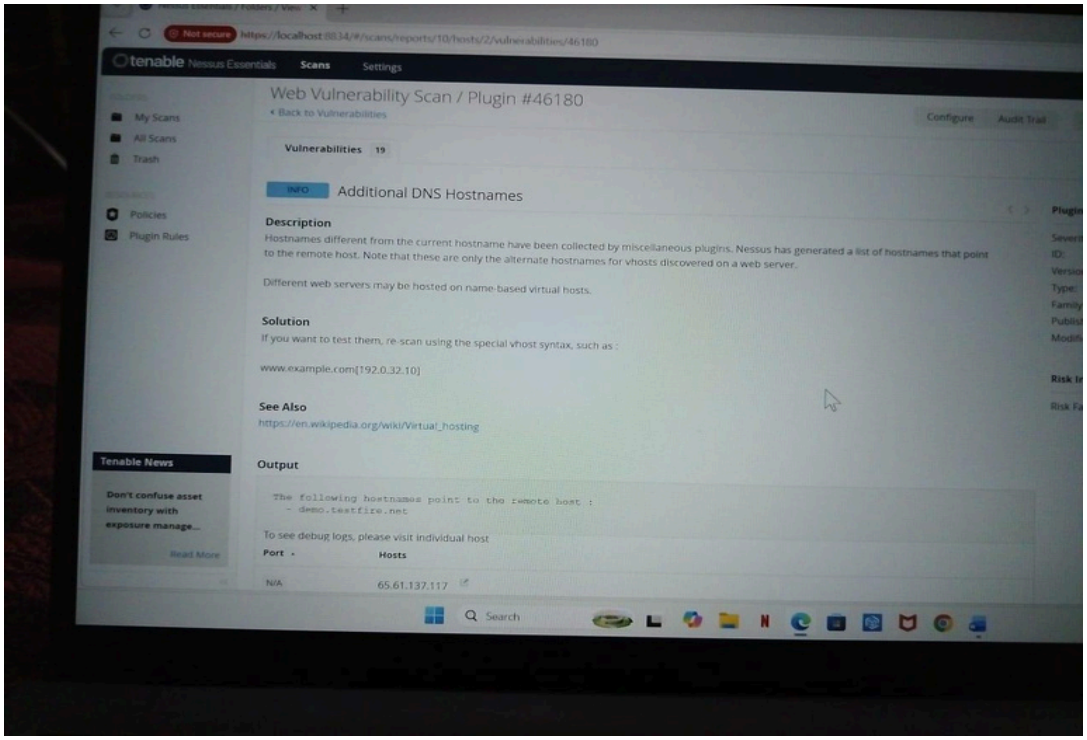
Targeted attacks on weaker endpoints

Vulnerability Path:

https://localhost (as identified in Nessus scan)

Steps to Reproduce:

- Perform a vulnerability scan using Nessus on the target system.
- Navigate to the “Additional DNS Hostnames” section in the results.
- Observe the list of alternate hostnames pointing to the same server.
- Verify that multiple hostnames resolve to the target IP address.



Recommendation:

•Remove unused or unnecessary DNS hostnames.

•Restrict access to internal or hidden subdomains.

•Implement proper access control for all hostnames.

•Regularly audit DNS configurations and records.

8. Vulnerability Name: Nessus SYN Scanner (Open Ports Detected)

CWE: 200 (Information Exposure)

Description:

The Nessus SYN scan identified open ports on the target system (e.g., port 80 and 443). SYN scanning is a stealthy technique used to detect open ports without completing the full TCP handshake.

Open ports indicate exposed services running on the system. If these services are not properly secured, they can be targeted by attackers for exploitation. This information is valuable during the reconnaissance phase of an attack.

Business Impact:

Exposure of running services to attackers

Increased attack surface

Possibility of exploitation of vulnerable services

Unauthorized access or service compromise

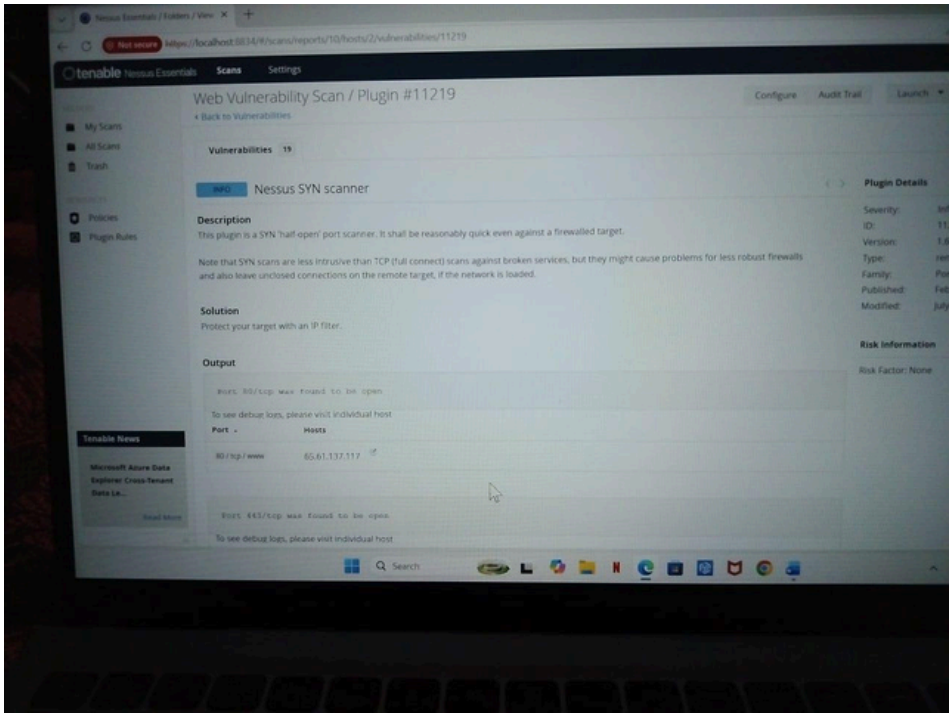
Vulnerability Path:

Target Host: 65.61.137.117

Ports: 80 (HTTP), 443 (HTTPS)

Steps to Reproduce:

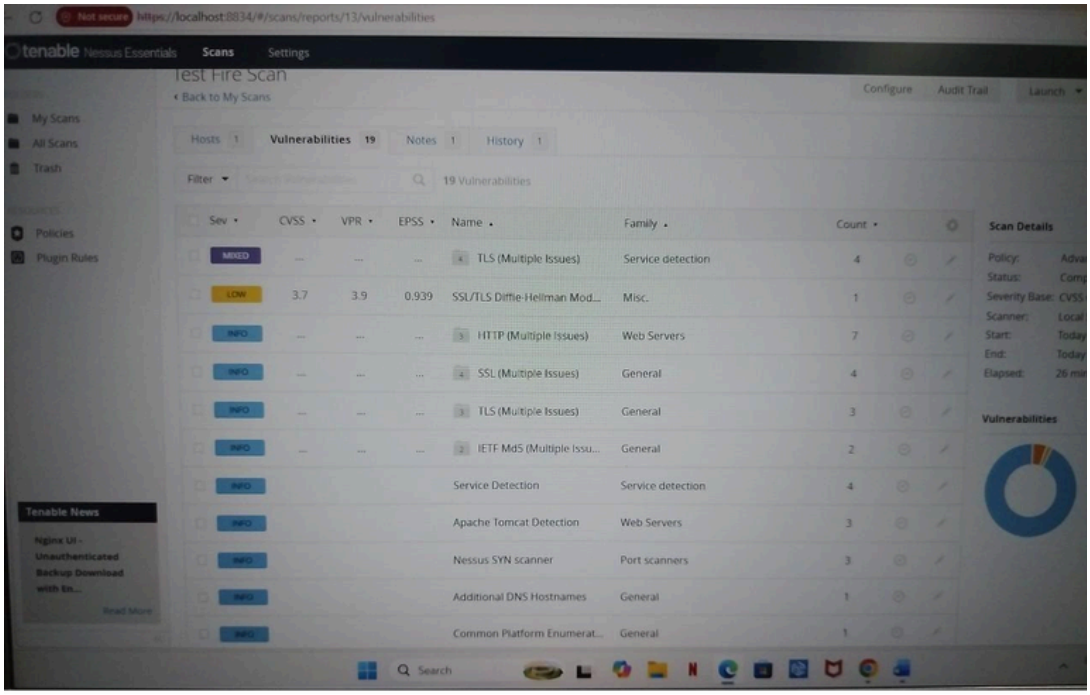
- Perform a Nessus vulnerability scan using SYN scan technique
- Navigate to “Nessus SYN Scanner” results
- Observe detected open ports (e.g., 80, 443)
- Verify services running on these ports

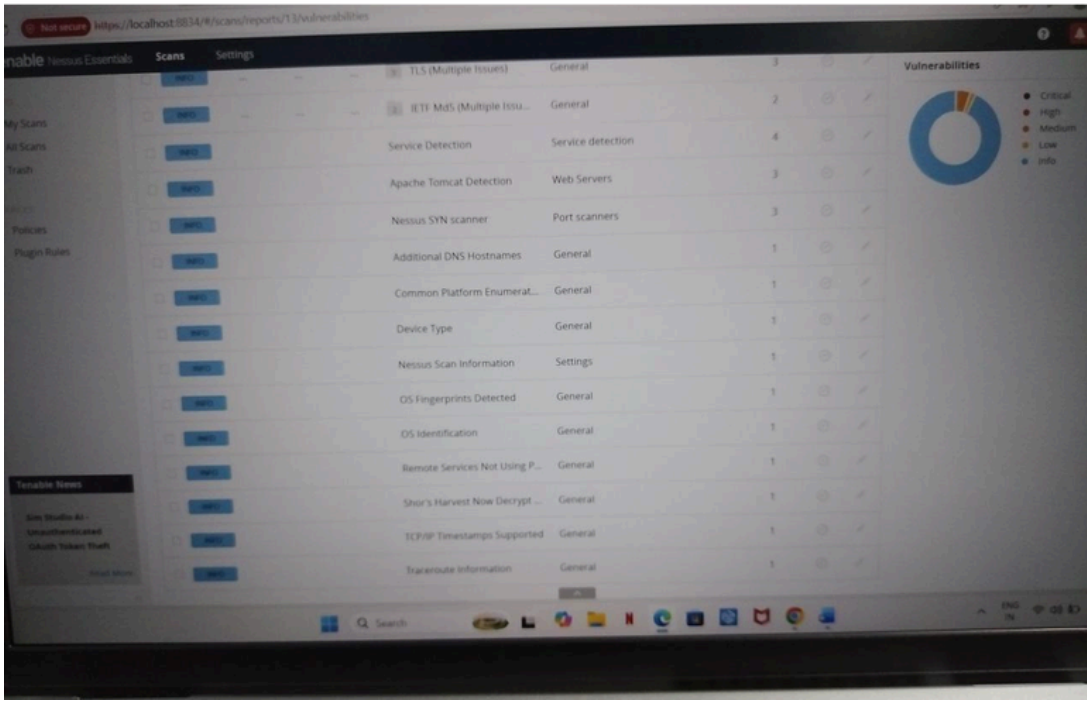


Recommendation:

•Closeunused ports onthesystem •Restrict access using firewall rules •Use IP filtering to allow only trusted sources •Ensure all exposed services are securely configured and updated •Implement intrusion detection/prevention systems (IDS/IPS)

NESSUS SCAN REPORT:





Stage-2

Vulnerability Analysis&RiskAssessment

Overview:

In this stage, the vulnerabilities identified using Nessus are analyzed based on their severity, associated plugins, and potential impact. The analysis helps in understanding the risks and prioritizing mitigation strategies to enhance system security.

📊 Detailed Analysis of 8 Vulnerabilities

1. TLS (Multiple Issues)

Severity: Medium

Analysis: The scan detected insecure or outdated TLS configurations such as older protocol versions (TLS 1.0/1.1).

Risk: Weak encryption can allow attackers to exploit vulnerabilities and decrypt sensitive data.

2. SSL (Multiple Issues)

Severity: Medium

Analysis: SSL configuration issues include weak certificates, outdated hashing algorithms, or insecure cipher suites.

Risk:Susceptibilitytoattacks likeMITM (Man-in-the-Middle) and data tampering.

3. HTTP (Multiple Issues)

Severity: Medium

Analysis: The use of HTTP instead of HTTPS leads to unencrypted communication and exposure of sensitive data.

Risk:Data interceptionand sessionhijacking.

4. IETF MD5 (Multiple Issues)

Severity: Medium

Analysis: The system uses MD5 hashing algorithm, which is cryptographically weak and vulnerable to collision attacks.

Risk: Certificate forgery and data integrity compromise.

5. Service Detection

Severity: Info

Analysis: Nessus identified services running on open ports of the system.

Risk: Attackers can use this information to identify targets and exploit known vulnerabilities.

6. OS Identification

Severity: Info

Analysis: The operating system details of the target were identified through scanning techniques.

Risk: Helps attackers tailor specific exploits based on OS vulnerabilities.

7. Additional DNS Hostnames

Severity: Info

Analysis: Additional hostnames pointing to the same server were discovered.

Risk: Exposure of hidden services or subdomains increases attack surface.

8. Nessus SYN Scanner

Severity: Info

Analysis: SYN scan revealed open ports such as HTTP (80) and HTTPS (443).

Risk: Open ports expose services that can be targeted for attacks.

S. No.	Vulnerability	Severity	Risk Level	Key Impact
1.	TLS Issues	Medium	High	Weak encryption
2.	SSL Issues	Medium	High	MITM attacks
3.	HTTP Issues	Medium	High	Data exposure
4.	MD5 Issues	Medium	High	Weak hashing
5.	Service Detection	Info	Medium	Reconnaissance
6.	OS Identification	Info	Medium	Targeted attacks
7.	DNS Hostnames	Info	Medium	Attack surface increase
8.	SYN Scanner	Info	Medium	Open port exposure
8.	SYN Scanner	Info	Medium	Open port exposure

Stage-3

🟦 TITLE

Implementation of Zero Trust Network Access (ZTNA) using Zscaler for Secure User-to-Application Connectivity

🟦 OVERVIEW

In this stage, a Zero Trust Network Access (ZTNA) model is implemented using Zscaler to enhance system security by providing secure, identity-based access to applications. Unlike traditional security approaches such as Virtual Private Networks (VPNs), which grant full network access after authentication, ZTNA follows the principle of “Never Trust, Always Verify.” This ensures that users are continuously validated before accessing any application.

Access decisions are not based solely on usernames and passwords but also consider multiple real-time factors such as user identity, device posture, and access policies defined within the Zscaler platform. By using a cloud-based architecture, Zscaler ensures that internal applications remain hidden from the internet, thereby reducing the attack surface and preventing unauthorized access.

The implementation of ZTNA in this project focuses on providing application-specific access rather than network-level access. Each user is granted access only to the required application based on predefined policies, ensuring least privilege access. This approach improves security, enhances user experience, and provides detailed logging and monitoring for all user activities.

🔐 KEY COMPONENTS OF STAGE 3

🔹 1. User Identity Verification

User authentication is performed using an Identity Provider such as Azure Active Directory integrated with Zscaler. Multi-Factor Authentication (MFA) is enforced to ensure secure login, and user roles and group memberships are verified before granting access.

👉 Ensures that only authorized and verified users can access specific applications

🔹 2. Device Posture Assessment

Before granting access, Zscaler checks the security posture of the user's device. This includes verifying whether the device has updated antivirus software, a secure operating system, and proper security configurations.

👉 Prevents compromised or unsafe devices from accessing internal applications

🔹 3. Application-Level Access Control

Access is granted only to specific applications rather than the entire network. Zscaler creates a secure micro-tunnel between the user and the application, ensuring that the internal network remains invisible.

👉 Eliminates lateral movement and reduces attack surface

🔹 4. Policy-Based Access Control

Zscaler enforces access policies based on user identity, application type, and security requirements. These policies ensure that users can only access resources that are necessary for their role.

👉 Ensures least privilege access and better control

🔹 5. Continuous Monitoring and Logging

All user activities are continuously monitored and logged within the Zscaler platform. Any suspicious behavior or anomaly is detected in real time, and access can be revoked immediately.

👉 Improves threat detection and incident response

CONCLUSION

Stage 1 Conclusion

Stage 1 focused on identifying and analyzing common web application vulnerabilities based on OWASP standards. Various security issues such as improper input validation, missing security headers, and outdated protocols were studied in detail. These vulnerabilities highlight how weak security practices can expose applications to serious threats like data breaches and unauthorized access. By understanding these vulnerabilities and their impacts, effective mitigation strategies can be applied to strengthen application security.

Stage 2 Conclusion

Stage 2 involved performing vulnerability assessment using the Nessus tool. The scan results provided detailed insights into existing security weaknesses, including their severity levels, potential risks, and recommended solutions. This stage helped in understanding real-world vulnerabilities and how automated tools can be used to detect them efficiently. The analysis of these vulnerabilities emphasizes the importance of regular security testing and timely remediation to maintain a secure system.

Stage 3 Conclusion

Stage 3 introduced the implementation of Zero Trust Network Access (ZTNA) using Zscaler. This approach ensures that access to applications is granted based on strict identity verification and security policies rather than network location. By providing application-specific access and continuous monitoring, ZTNA significantly reduces the attack surface and prevents unauthorized access. This stage demonstrates how modern security frameworks can replace traditional models to achieve a more secure and scalable system.

FUTURE SCOPE

Stage 1 Future Scope

Future improvements can include the use of advanced automated tools for detecting vulnerabilities more efficiently and accurately. Integration with secure coding practices and real-time vulnerability monitoring systems can further enhance application security. Additionally, implementing developer training programs on secure coding standards can help prevent vulnerabilities at the development stage itself.

Stage 2 Future Scope

In the future, vulnerability scanning can be integrated with continuous security monitoring systems to provide real-time threat detection. Advanced tools with AI and machine learning capabilities can be used to predict potential vulnerabilities before they are exploited. Automation of patch management and remediation processes can also improve efficiency and reduce manual effort in maintaining system security.

Stage 3 Future Scope

The Zero Trust model can be further enhanced by integrating advanced technologies such as artificial intelligence for behavioral analysis and threat prediction. Expanding ZTNA implementation to cloud environments and hybrid networks can improve scalability and flexibility. Additionally, incorporating real-time analytics and adaptive access control mechanisms can further strengthen security and improve user experience.